

# Topology and Manifolds

MH3600

## Chapter 2: Topological Spaces

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# Outline

- 1 Topological spaces
- 2 Basis of a topology
- 3 The order topology
- 4 The product topology
- 5 The subspace topology
- 6 Closed sets and limit points
- 7 References

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- 2 Basis of a topology
- 3 The order topology
- 4 The product topology
- 5 The subspace topology
- 6 Closed sets and limit points
- 7 References

# Topological spaces

The following definition may seem a bit abstract, but as we work through the various ways of constructing topological spaces, we will get a better feeling for what the concept means.

## Definition

A **topology** on a set  $X$  is a collection  $\mathcal{T}$  of subsets of  $X$  having the following properties:

- (i)  $\emptyset$  and  $X$  are in  $\mathcal{T}$ .
- (ii) The **union** of the elements of **any** subcollection of  $\mathcal{T}$  is in  $\mathcal{T}$ .
- (iii) The **intersection** of the elements of any **finite** subcollection of  $\mathcal{T}$  is in  $\mathcal{T}$ .

A set  $X$  for which a topology  $\mathcal{T}$  has been specified is called a **topological space**. Properly speaking, a **topological space** is the ordered pair  $(X, \mathcal{T})$  but we often omit specific mention of  $\mathcal{T}$  if no confusion will arise.

# Open sets

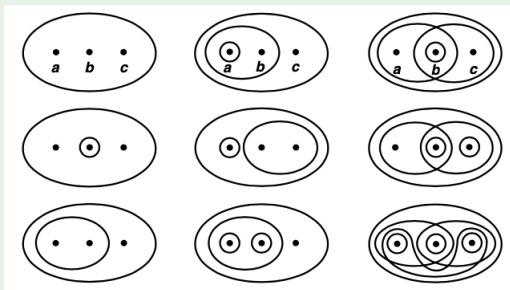
## Definition

Let  $(X, \mathcal{T})$  a topological space. We say that a subset  $U$  of  $X$  is an **open set** of  $X$  if  $U$  belongs to the collection  $\mathcal{T}$ .

## Example

Let  $X$  be a three-element set,  $X = \{a, b, c\}$ . Nine topologies:

Taken from [Munkres]



## Example

Let  $X$  be a set. The **power set** (collection of all subsets of  $X$ ) is a topology on  $X$  called the **discrete topology**. The collection consisting of  $X$  and  $\emptyset$  only is also a topology on  $X$  called the **indiscrete or trivial topology**.

# Comparison of topologies

## Definition

Suppose that  $\mathcal{T}$  and  $\mathcal{T}'$  are two topologies on a given set  $X$ . If  $\mathcal{T}' \supset \mathcal{T}$ , we say that  $\mathcal{T}'$  is **finer/larger/stronger** than  $\mathcal{T}$ ; if  $\mathcal{T}'$  properly contains  $\mathcal{T}$ , we say that  $\mathcal{T}'$  is **strictly finer/larger/stronger** than  $\mathcal{T}$ . We also say that  $\mathcal{T}$  is **coarser/smaller/weaker** than  $\mathcal{T}'$ , or **strictly coarser/smaller/weaker**, in these two respective situations. We say  $\mathcal{T}$  is **comparable** with  $\mathcal{T}'$  if either  $\mathcal{T}' \supset \mathcal{T}$  or  $\mathcal{T} \supset \mathcal{T}'$ .

Two topologies on  $X$  need not be comparable. In the preceding Figure, the topology in the upper right-hand corner is strictly finer than each of the three topologies in the first column and strictly coarser than each of the other topologies in the third column. But it is not comparable with any of the topologies in the second column.

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- 1 Topological spaces
- 2 Basis of a topology**
- 3 The order topology
- 4 The product topology
- 5 The subspace topology
- 6 Closed sets and limit points
- 7 References

# Bases of a topology

There are three main ways to **define topologies** on a set  $X$ :

- Giving the **full list** of open sets like in the examples above. Usually very difficult.
- **New** topologies out of **old ones**: **subspaces, quotients, and products**. Will come later.
- Specify a **smaller collection of subsets** of  $X$  and defines the topology in terms of that. This is what we do now using the bases of a topology.

## Definition

$X$  a set, a **basis** for a topology on  $X$  is a collection  $\mathcal{B}$  of subsets of  $X$  (called **basis elements**) such that

- For each  $x \in X$ , there is at least one basis element  $B$  containing  $x$ .
- If  $x$  belongs to the intersection of two basis elements  $B_1$  and  $B_2$ , then there is a basis element  $B_3$  containing  $x$  such that  $B_3 \subset B_1 \cap B_2$ .

## Topology $\mathcal{T}$ generated by a basis

If  $\mathcal{B}$  satisfies these two conditions, we define the **topology  $\mathcal{T}$  generated by  $\mathcal{B}$**  as follows: A subset  $U$  of  $X$  is said to be open in  $X$  (that is, to be an element of  $\mathcal{T}$ ) if for each  $x \in U$ , there is a basis element  $B \in \mathcal{B}$  such that  $x \in B$  and  $B \subset U$ . Note that each basis element is itself an element of  $\mathcal{T}$ . We check later shortly that the collection  $\mathcal{T}$  is indeed a topology on  $X$ .

# Examples

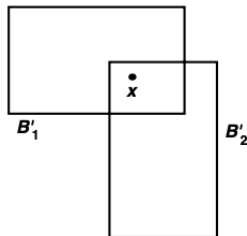
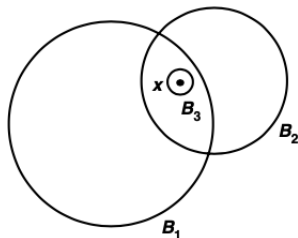
## Example

Let  $\mathcal{B}$  be the collection of all **circular regions** (interiors of circles) in the plane. Then  $\mathcal{B}$  satisfies both conditions for a basis. The second condition is illustrated in the figure. In the topology generated by  $\mathcal{B}$ , a subset  $U$  of the plane is open if every  $x$  in  $U$  lies in some circular region contained in  $U$ .

## Example

Let  $\mathcal{B}'$  be the collection of all **rectangular regions** (interiors of rectangles) in the plane, where the rectangles have sides parallel to the coordinate axes. Then  $\mathcal{B}'$  satisfies both conditions for a basis. The second condition is trivial, because the intersection of any two basis elements is itself a basis element (or empty). As we shall see later, the basis  $\mathcal{B}'$  generates the same topology on the plane as the basis  $\mathcal{B}$  given in the preceding example.

Taken from [Munkres]



## Example

If  $X$  is any set, the collection of all one-point subsets of  $X$  is a basis for the **discrete topology** on  $X$ .

## Theorem

Let  $\mathcal{B}$  be a *basis for a topology* on  $X$ . Define

$$\mathcal{T} = \{U \subset X \mid \text{for all } x \in U, \text{ there is } B \in \mathcal{B} \text{ such that } x \in B \subset U\},$$

the *“topology” generated by  $\mathcal{B}$* . Then  $\mathcal{T}$  is a topology on  $X$ .

## Proof

We consider the definition of “topology.”

- (i)  $\emptyset \in \mathcal{T}$  vacuously. Now  $X \in \mathcal{T}$  since each  $x \in X$  satisfies  $x \in B \subset X$  for some  $B \in \mathcal{B}$  by the definition of topology generated by  $\mathcal{B}$ .
- (ii) Let  $\{U_\alpha\}_{\alpha \in J}$  be an arbitrary collection of elements of  $\mathcal{T}$ . Let  $U = \bigcup_{\alpha \in J} U_\alpha$ . For  $x \in U$  we have  $x \in U_\alpha$  for some  $\alpha \in J$ . Since  $U_\alpha \in \mathcal{T}$ , then by the definition of “topology generated by  $\mathcal{B}$ ,”  $x \in B \subset U_\alpha$  for some  $B \in \mathcal{B}$ . So  $x \in B \subset U$  and hence by definition  $U$  is open.

## Proof (continued)

(iii) Let  $U_1, U_2 \in \mathcal{T}$ . For  $x \in U_1 \cap U_2$ , by the definition of “topology generated by  $\mathcal{B}$ ”, there is  $B_1 \subset U_1$  and  $B_2 \subset U_2$  with  $B_1, B_2 \in \mathcal{B}$  and  $x \in B_1, x \in B_2$ . By part (2) of the definition of “basis for a topology”, there is  $B_3 \in \mathcal{B}$  with  $x \in B_3$  and  $B_3 \subset B_1 \cap B_2 \subset U_1 \cap U_2$ . Hence  $U_1 \cap U_2 \in \mathcal{T}$ . Next, by mathematical induction, any finite collection  $\{U_1, U_2, \dots, U_n\} \subset \mathcal{T}$  satisfies  $U_1 \cap U_2 \cap \dots \cap U_n \in \mathcal{T}$ .

So  $\mathcal{T}$  satisfies the definition of topology and  $\mathcal{T}$  is a topology on  $X$ . ■

Another way of describing the topology generated by a basis is given in the following lemma:

### Lemma

*Let  $X$  be a set and let  $\mathcal{B}$  be a basis for a topology  $\mathcal{T}$  on  $X$ . Then  $\mathcal{T}$  equals the collection of all unions of elements of  $\mathcal{B}$ .*

### Proof

As stated in the Theorem above, all elements of  $\mathcal{B}$  are open and so in  $\mathcal{T}$ . Since  $\mathcal{T}$  is a topology, then by part (ii) of the definition of “topology”, any union of elements of  $\mathcal{B}$  are in  $\mathcal{T}$ . So  $\mathcal{T}$  contains all unions of elements of  $\mathcal{B}$ .

Next, suppose  $U \in \mathcal{T}$ . For each  $x \in U$  choose  $B_x \in \mathcal{B}$  such that  $x \in B_x \subset U$  (which can be done by the definition of “topology  $\mathcal{T}$  generated by  $\mathcal{B}$ ”). Then  $U = \bigcup_{x \in U} B_x$ , so  $U$  equals a union of elements of  $\mathcal{B}$ . Since  $U$  is an arbitrary element of  $\mathcal{T}$ , then all elements of  $\mathcal{T}$  are unions of elements of  $\mathcal{B}$  and the result follows. ■

- This lemma states that every open set  $U$  in  $X$  can be expressed as a union of basis elements. **This expression for  $U$  is not unique.** Thus the use of the term “basis” in topology differs from its use in linear algebra, where the expression of a given vector as a linear combination of basis vectors is unique.
- We have described in two different ways how to go from a basis to the topology it generates. Sometimes **we need to go in the reverse direction, from a topology to a basis generating it.** The next lemma is one way of obtaining a basis for a given topology.

## Lemma

Let  $(X, \mathcal{T})$  be a topological space. Suppose that  $\mathcal{C}$  is a collection of open sets of  $X$  such that for each open subset  $U \subset X$  and each  $x \in U$ , there is an element  $C \in \mathcal{C}$  such that  $x \in C \subset U$ . Then  $\mathcal{C}$  is a basis for the topology  $\mathcal{T}$  on  $X$ .

Recall:  $\mathcal{B}$  basis if:

- (i) For each  $x \in X$ , there is  $B \in \mathcal{B}$  containing  $x$ .
- (ii) If  $x \in B_1 \cap B_2$  with  $B_1, B_2 \in \mathcal{B}$ , then there is  $B_3 \in \mathcal{B}$  such that  $x \in B_3 \subset B_1 \cap B_2$ .

## Proof

First we show that  $\mathcal{C}$  is a basis.

For the first part of the definition of basis, for  $x \in X$  (since  $X$  itself is an open set) then, by hypothesis on  $\mathcal{C}$ , there is  $C \in \mathcal{C}$  such that  $x \in C \subset X$ .

For the second part of the definition of basis, let  $x \in C_1 \cap C_2$  where  $C_1, C_2 \in \mathcal{C}$ . Since  $C_1$  and  $C_2$  are open then so is  $C_1 \cap C_2$  by part (3) of the definition of topology. Then, by hypothesis on  $\mathcal{C}$ , there is  $C_3 \in \mathcal{C}$  such that  $x \in C_3 \subset C_1 \cap C_2$ . So both parts of the definition of basis are satisfied and  $\mathcal{C}$  is a basis for a topology on  $X$ .

So far:  $\mathcal{C}$  is made from open sets in  $\mathcal{T}$  and is a basis for a topology.

Recall: to  $\mathcal{B}$  we associate  $\mathcal{T} = \{U \subset X \mid \text{for all } x \in U, \text{ there is } B \in \mathcal{B} \text{ such that } x \in B \subset U\}$ .

## Proof (continued)

Let  $\mathcal{T}'$  be the topology on  $X$  generated by  $\mathcal{C}$  (we now show that  $\mathcal{T} = \mathcal{T}'$ ). First, if  $U \in \mathcal{T}$  and  $x \in U$ , then by hypothesis on  $\mathcal{C}$ , there is  $C \in \mathcal{C}$  such that  $x \in C \subset U$ . So, by the definition of “topology generated by”  $\mathcal{C}$ ,  $U$  is in  $\mathcal{T}'$  and hence  $\mathcal{T} \subset \mathcal{T}'$ .

Conversely, if  $W$  belongs to  $\mathcal{T}'$  then  $W$  is a union of elements of  $\mathcal{C}$  by the previous lemma. Now each element of  $\mathcal{C}$  is an element of  $\mathcal{T}$  (see the comment at the end of the definition of “topology generated by”) and a union of open sets is open (definition of topology, part **(ii)**) so  $W$  belongs to  $\mathcal{T}$ . That is,  $\mathcal{T}' \subset \mathcal{T}$ . Therefore,  $\mathcal{T} = \mathcal{T}'$ . ■

# Comparison of topologies using bases

When **topologies are given by bases**, it is useful to have a criterion in terms of the bases elements for **determining whether one topology is finer than another**.

## Lemma

Let  $\mathcal{B}$  and  $\mathcal{B}'$  be bases for topologies  $\mathcal{T}$  and  $\mathcal{T}'$ , respectively, on  $X$ . *Then the following are equivalent:*

- (i)  $\mathcal{T}'$  is finer than  $\mathcal{T}$  ( $\mathcal{T}' \supset \mathcal{T}$ ).
- (ii) For each  $x \in X$  and each basis element  $B \in \mathcal{B}$  containing  $x$ , there is a basis element  $B' \in \mathcal{B}'$  such that  $x \in B' \subset B$ .

## Proof

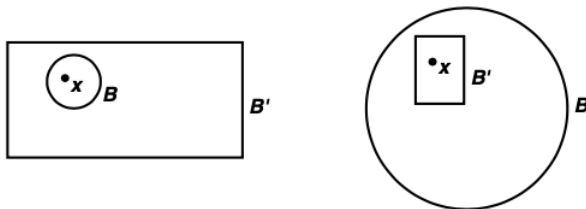
**(ii)  $\Rightarrow$  (i)** Given  $U \in \mathcal{T}$ , let  $x \in U$ . Since  $\mathcal{B}$  generates  $\mathcal{T}$ , there is  $B \in \mathcal{B}$  such that  $x \in B \subset U$ . By hypothesis **(ii)**, there is  $B' \in \mathcal{B}'$  such that  $x \in B' \subset B$ . Then  $x \in B' \subset U$  and by the definition of “topology generated by”  $\mathcal{B}'$ , we have  $U \in \mathcal{T}'$ . So  $\mathcal{T} \subset \mathcal{T}'$  and **(i)** follows.

**(i)  $\Rightarrow$  (ii)** Let  $x \in X$  and  $B \in \mathcal{B}$  where  $x \in B$ . Since  $\mathcal{B}$  generates  $\mathcal{T}$ , then  $B \in \mathcal{T}$ . By hypothesis **(i)**,  $\mathcal{T} \subset \mathcal{T}'$  and so  $B \in \mathcal{T}'$ . Since  $\mathcal{T}'$  is generated by  $\mathcal{B}'$ , there is (by definition)  $B' \in \mathcal{B}'$  such that  $x \in B' \subset B$  and **(ii)** follows. ■

## Example

One can now see that the collection  $\mathcal{B}$  of all circular regions in the plane generates the same topology as the collection  $\mathcal{B}'$  of all rectangular regions. The figure illustrates the proof.

Taken from [Munkres]



# Subbasis of a topology

## Definition

A **subbasis** for a topology on  $X$  is a collection of subsets of  $X$  whose union equals  $X$ . The **topology generated by the subbasis**  $\mathcal{S}$  is defined to be the collection  $\mathcal{T}$  of all unions of finite intersections of elements of  $\mathcal{S}$ .

## Theorem

Let  $\mathcal{S}$  be a subbasis for a topology on  $X$ . Define  $\mathcal{T}$  to be *all unions of finite intersections of elements of  $\mathcal{S}$* . Then  $\mathcal{T}$  is a topology on  $X$ .

## Proof

Let  $\mathcal{B}$  be the set of all finite intersections of elements of  $\mathcal{S}$  :

$$\mathcal{B} = \{S_1 \cap S_2 \cap \cdots \cap S_n \mid n \in \mathbb{N}; S_1, S_2, \dots, S_n \in \mathcal{S}\}$$

Let us show that  $\mathcal{B}$  is a basis. Let  $x \in X$ . Then  $x \in S$  for some  $S \in \mathcal{S}$  by the definition of subbasis, and so  $x \in S$  where  $S \in \mathcal{B}$ . So part (1) of the definition of “ $\mathcal{B}$  is a basis” is satisfied. Now let  $B_1, B_2 \in \mathcal{B}$ . Then  $B_1 = S_1 \cap S_2 \cap \cdots \cap S_m$  and  $B_2 = S'_1 \cap S'_2 \cap \cdots \cap S'_n \in \mathcal{B}$  and  $B_1 \cap B_2$  also belongs to  $\mathcal{B}$  as it is a finite intersection of elements in  $\mathcal{S}$  so that part (2) of the definition of “ $\mathcal{B}$  is a basis” is satisfied and so  $\mathcal{B}$  is a basis for a topology on  $X$ . The topology for which  $\mathcal{B}$  is a basis is, by Lemma on slide 15, the topology consisting of all unions of elements of  $\mathcal{B}$ . This is precisely the collection of sets in  $\mathcal{T}$ . So  $\mathcal{T}$  is a topology on  $X$ . ■

# Outline

- 1 Topological spaces
- 2 Basis of a topology
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- 7 References

# The order topology

## Definition

Let  $X$  be a set with an **order relation**  $<$ . The following sets are **intervals** in  $X$ :

$$(a, b) = \{x \in X \mid a < x < b\} \text{ (open intervals)}$$

$$(a, b] = \{x \in X \mid a < x \leq b\} \text{ (half-open intervals)}$$

$$[a, b) = \{x \in X \mid a \leq x < b\} \text{ (half-open intervals)}$$

$$[a, b] = \{x \in X \mid a \leq x \leq b\} \text{ (closed intervals)}$$

## Definition

Let  $X$  be a set with a simple order relation and assume  $X$  has more than one element. Let  $\mathcal{B}$  be the collection of all sets of the following types:

- (1) All open intervals  $(a, b)$  in  $X$ .
- (2) All intervals of the form  $[a_0, b)$  where  $a_0$  is the least element (if one exists) of  $X$ .
- (3) All intervals of the form  $(a, b_0]$  where  $b_0$  is the greatest element (if one exists) of  $X$ .

The collection  $\mathcal{B}$  is a basis for a topology on  $X$  called the order topology.

Of course we must verify that  $\mathcal{B}$  really is a basis for a topology.

## Theorem

Let  $X$  be a set with an order relation and let  $\mathcal{B}$  consist of all open intervals  $(a, b)$ , all intervals  $[a_0, b)$ , and all intervals  $(a, b_0]$ , where  $a_0$  is the least element of  $X$  and  $b_0$  is the greatest element of  $X$  (if such exist). Then  $\mathcal{B}$  is a basis for a topology on  $X$ .

## Proof

We confirm that the definition of basis of a topology is satisfied. If  $a_0$  is the least element of  $X$  then  $a_0 \in [a_0, b)$  for all  $b \neq a_0$ . If  $b_0$  is the greatest element of  $X$  then  $b_0 \in (a, b_0]$  for all  $a \neq b_0$ . If  $x \in X$  is neither the least nor greatest element of  $X$  then  $x \in (a, b)$  for some  $a, b \in X$ . So part **(i)** of the definition is satisfied. For part **(ii)** of the definition, let  $B_1, B_2 \in \mathcal{B}$ . It can be checked that  $B_1 \cap B_2 \in \mathcal{B}$  by checking the several cases that can arise. So part **(ii)** is satisfied and  $\mathcal{B}$  is in fact a basis for a topology on  $X$ . ■

# Examples order topology

## Example

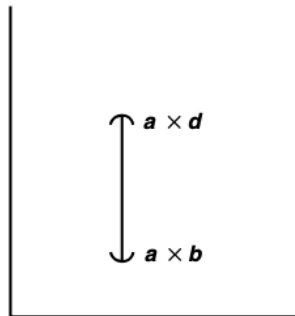
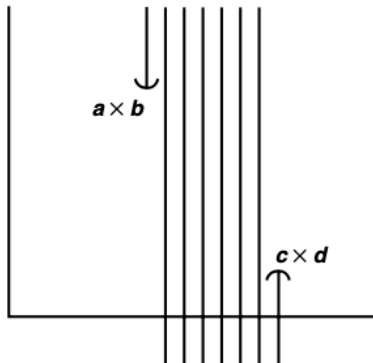
The **standard topology** on  $\mathbb{R}$  is just the order topology derived from the usual order on  $\mathbb{R}$ .

## Example

Consider the set  $\mathbb{R} \times \mathbb{R}$  in the **dictionary/lexicographic order** (see textbook Chapter 1):  $(a, b) < (c, d)$  means  $a < c$  or  $a = c, b < d$ .

The set  $\mathbb{R} \times \mathbb{R}$  has neither a largest nor a smallest element, so the order topology on  $\mathbb{R} \times \mathbb{R}$  has as basis the collection of all open intervals of the form  $((a, b), (c, d))$  for  $a < c$ , and for  $a = c$  and  $b < d$ . The subcollection consisting of only intervals of the second type is also a basis for the order topology on  $\mathbb{R} \times \mathbb{R}$ , as you can check.

Taken from [Munkres]



## Example

Let  $X = \{1, 2\} \times \mathbb{Z}_+$  with the dictionary order. Then  $(1, 1)$  is the smallest element of  $X$ , though there is no greatest element of  $X$ . The ordering produces the inequalities:  $(1, 1) < (1, 2) < (1, 3) < \cdots < (2, 1) < (2, 2) < \cdots$  where the first " $\cdots$ " indicates that all elements of the form  $(1, n)$  are present. Notice that all but one singleton is in the basis  $\mathcal{B}$  for the order topology:

$$(1, 1) = [(1, 1), (1, 2))$$

$$(1, n) = ((1, n-1), (1, n+1)) \text{ for } n > 1$$

$$(2, n) = ((2, n-1), (2, n+1)) \text{ for } n > 1$$

but a basis element containing  $(2, 1)$  must be of the form  $((a_1, a_2), (b_1, b_2))$  where  $(a_1, a_2) < (2, 1)$  and  $(2, 1) < (b_1, b_2)$ . But then  $(a_1, a_2)$  is of the form  $(1, n)$  for some  $n \in \mathbb{Z}_+$ , so the interval  $((a_1, a_2), (b_1, b_2))$  contains an infinite number of elements of  $X$  less than  $(2, 1)$ . Now any open set containing  $(2, 1)$  must contain a basis element about  $(2, 1)$  and so the singleton  $(2, 1)$  cannot be open. It is the only singleton in the topological space which is not open.

In particular, this implies that **the order topology on  $X$  is not the discrete topology**.

# Rays

## Definition

If  $X$  is a set with an order relation  $<$ , and  $a \in X$  then there are four subsets of  $X$ , called **rays determined by  $a$** . They are the following:

$$(a, +\infty) = \{x \in X \mid x > a\}$$

$$(-\infty, a) = \{x \in X \mid x < a\}$$

$$[a, +\infty) = \{x \in X \mid x \geq a\}$$

$$(-\infty, a] = \{x \in X \mid x \leq a\}$$

The first two types of rays are called **open rays** and the last two types are called **closed rays**.

( $\pm\infty$  are not elements of  $X$ , they appear as a notation).

The open rays in  $X$  are open sets in the order topology since:

- If  $X$  has a greatest element  $b_0$  then  $(a, +\infty) = (a, b_0] \in \mathcal{B}$  is open.
- If  $X$  does not have a greatest element then  $(a, +\infty) = \cup_{x>a}(a, x)$  is open.
- If  $X$  has a smallest element  $a_0$  then  $(-\infty, a) = [a_0, a) \in \mathcal{B}$  is open.
- If  $X$  does not have a smallest element then  $(-\infty, a) = \cup_{x<a}(x, a)$  is open.

## Theorem

Let  $X$  be a set with an order relation. The *open rays form a subbasis* for the order topology  $\mathcal{T}$  on  $X$ .

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# The product topology

If  $X$  and  $Y$  are topological spaces, then there is a **natural topology** on the **Cartesian product** set  $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$  that we call the **product topology**.

## Definition

Let  $X$  and  $Y$  be topological spaces. The **product topology** on set  $X \times Y$  is the topology having as basis the collection  $\mathcal{B}$  of all sets of the form  $U \times V$ , where  $U$  is an open subset of  $X$  and  $V$  is an open subset of  $Y$ .

We need to **verify that  $\mathcal{B}$  is a basis or a topology**.

- Since  $X$  is open and  $Y$  is open, then  $X \times Y \in \mathcal{B}$  is open and part **(i)** of the definition of “basis” is satisfied.
- For part **(ii)** of the definition, let  $B_1 = U_1 \times V_1$  and  $B_2 = U_2 \times V_2$  be elements of  $\mathcal{B}$ , then  $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$  is a basis element because  $U_1 \cap U_2$  and  $V_1 \cap V_2$  are open in  $X$  and  $Y$ , respectively.

The following result tells us how to find a basis for the product topology on  $X \times Y$  in terms of bases of  $X$  and  $Y$ .

## Theorem

If  $\mathcal{B}$  is a *basis for the topology* of  $X$  and  $\mathcal{C}$  is a basis for the topology of  $Y$ , then the collection  $\mathcal{D} = \{B \times C \mid B \in \mathcal{B} \text{ and } C \in \mathcal{C}\}$  is a *basis for the topology of  $X \times Y$* .

## Proof

We recall this Lemma:

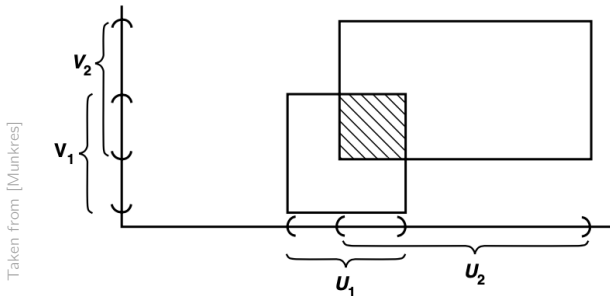
Let  $(X, \mathcal{T})$  be a topological space. Suppose that  $\mathcal{C}$  is a collection of open sets of  $X$  such that for each open subset  $U \subset X$  and each  $x \in U$ , there is an element  $C \in \mathcal{C}$  such that  $x \in C \subset U$ . Then  $\mathcal{C}$  is a basis for the topology  $\mathcal{T}$  on  $X$ .

Let  $W \subset X \times Y$  be an open set and let  $(x, y) \in W$ . By the definition of product topology, there is a basis element  $U \times V$ , where  $U$  is open in  $X$  and  $V$  is open in  $Y$ , such that  $(x, y) \in U \times V \subset W$ . So  $x \in U$  and  $y \in V$ . Since  $\mathcal{B}$  and  $\mathcal{C}$  are bases for  $X$  and  $Y$ , respectively, then there are open  $B \in \mathcal{B}$  and  $C \in \mathcal{C}$  such that  $x \in B \subset U$  and  $y \in C \subset V$ . Notice that  $B \times C$  is an element of the basis for the product topology and so is open and  $B \times C \in \mathcal{D}$ . That is,  $(x, y) \in B \times C \subset W$  where  $B \times C \in \mathcal{D}$ , so by the Lemma,  $\mathcal{D}$  is a basis for the product topology. ■

## Example

The standard topology on  $\mathbb{R}^2$  is the product topology on  $\mathbb{R} \times \mathbb{R}$  where we have the standard topology on  $\mathbb{R}$ . Since a basis for the standard topology on  $\mathbb{R}$  is  $\mathcal{B} = \{(a, b) \mid a, b \in \mathbb{R}, a < b\}$  (by the definition of “standard topology on  $\mathbb{R}$ ”), then the previous theorem implies that a basis for the standard topology on  $\mathbb{R} \times \mathbb{R}$  is

$$\mathcal{B}' = \{(a, b) \times (c, d) \mid a, b, c, d \in \mathbb{R}, a < b, c < d\}.$$



The following definition introduces projections which allow us to move sets from  $X \times Y$  into the constituent sets  $X$  and  $Y$ .

### Definition

Let  $\pi_1 : X \times Y \rightarrow X$  be defined by  $\pi_1((x, y)) = x$  and let  $\pi_2 : X \times Y \rightarrow Y$  be defined by  $\pi_2((x, y)) = y$ . The maps  $\pi_1$  and  $\pi_2$  are the **projections** of  $X \times Y$  onto its first and second factors, respectively.

For any  $U \subset X$ , we can consider the inverse image of  $U$  under  $\pi_1$  :

$$\begin{aligned}\pi_1^{-1}(U) &= \{(x, y) \in X \times Y \mid \pi_1((x, y)) \in U\} \\ &= \{(x, y) \in X \times Y \mid x \in U\} = U \times Y.\end{aligned}$$

**Important:** this involves inverse images and not inverse functions, so there is no concern about one to one-ness.) Similarly, for  $V \subset Y$  we have  $\pi_2^{-1}(V) = X \times V$ .

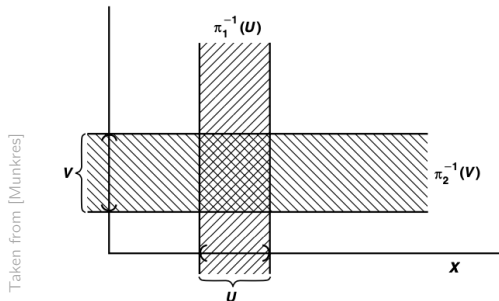
The inverse images of open sets in  $X$  and  $Y$  under projections can be used to create a subbasis for the product topology on  $X \times Y$  as follows.

## Theorem

The set

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ is open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ is open in } Y\}$$

*is a subbasis for the product topology on  $X \times Y$ .*



# Outline

- 1 Topological spaces
- 2 Basis of a topology
- 3 The order topology
- 4 The product topology
- 5 The subspace topology**
- 6 Closed sets and limit points
- 7 References

# The subspace topology

## Definition

Let  $X$  be a topological space with topology  $\mathcal{T}$ . If  $Y$  is a subset of  $X$ , then the set

$$\mathcal{T}_Y = \{Y \cap U \mid U \in \mathcal{T}\}$$

is a topology on  $Y$  called the **subspace topology**. With this topology,  $Y$  is called a **subspace** of  $X$ .

The previous definition claims that  $\mathcal{T}_Y$  is a topology. This is easy to confirm. First,  $\emptyset, Y \in \mathcal{T}_Y$  since  $\emptyset = Y \cap \emptyset$  and  $Y = Y \cap X$ . For finite intersections, we have

$$(U_1 \cap Y) \cap (U_2 \cap Y) \cap \cdots \cap (U_n \cap Y) = (U_1 \cap U_2 \cap \cdots \cap U_n) \cap Y \in \mathcal{T}_Y$$

and for arbitrary unions

$$\bigcup_{\alpha \in J} (U_\alpha \cap Y) = (\bigcup_{\alpha \in J} U_\alpha) \cap Y \in \mathcal{T}_Y.$$

# Bases for the subspace topology

Now we give a method by which to find a **basis for the subspace topology**.

## Lemma

*If  $\mathcal{B}$  is a basis for the topology of  $X$  and  $Y \subset X$  then the set  $\mathcal{B}_Y = \{B \cap Y \mid B \in \mathcal{B}\}$  is a basis for the subspace topology on  $Y$ .*

## Proof

Let  $V$  be open in  $Y$ . There is  $U$  open in  $X$  so that  $V = U \cap Y$ . Let  $y \in V = U \cap Y$ . Then since  $\mathcal{B}$  is a basis for the topology of  $X$ , there is (open)  $B \in \mathcal{B}$  such that  $y \in B \subset U$ . Then  $y \in B \cap Y \subset U \cap Y$ . Then by the same Lemma as earlier (slide 17),  $\mathcal{B}_Y$  is a basis for the subspace topology on  $Y$ . ■

## Lemma

Let  $Y$  be a subspace of  $X$ . If  $U$  is *open* in  $Y$  and  $Y$  is open in  $X$ , then  $U$  is *open* in  $X$ .

## Proof

Let  $U$  be open in  $Y$ . Then  $U = Y \cap V$  for some set  $V$  open in  $X$ . Since  $Y$  and  $V$  are both open in  $X$ , then  $Y \cap V = U$  is open in  $X$ . ■

## Lemma

If  $A$  is a subspace of  $X$  and  $B$  is a subspace of  $Y$ , then the *product topology* on  $A \times B$  is the *same as the topology*  $A \times B$  inherits *as a subspace* of  $X \times Y$ .

## Proof

Let  $U \times V$  be a basis element for the product topology on  $X \times Y$ . Then  $(U \times V) \cap (A \times B)$  is a basis element for the *subspace topology* on  $A \times B$  (Lemma slide 41). Now  $(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B)$ . Since  $U \cap A$  and  $V \cap B$  are open relative to  $A$  and  $B$ , respectively, then  $(U \cap A) \times (V \cap B)$  is a basis element for the *product topology* on  $A \times B$  (Definition slide 34). So the basis for the subspace topology on  $A \times B$  is a subset of the basis for the product topology on  $A \times B$ . Conversely, a basis element for the product topology on  $A \times B$  is of the form  $(U \cap A) \times (V \cap B)$  where  $U$  and  $v$  are open in  $X$  and  $Y$ , respectively, and from the equality above, this is a basis element for the subspace topology on  $A \times B$ . So the basis for the product topology on  $A \times B$  is a subset of the basis for the subspace topology on  $A \times B$ . So the bases are the same and, as claimed, the topologies are the same. ■

# Outline

- 1 Topological spaces
- 2 Basis of a topology
- 3 The order topology
- 4 The product topology
- 5 The subspace topology
- 6 Closed sets and limit points**
- 7 References

## Closed sets

We finally define **closed sets**. We also introduce several important topological concepts, such as **limit points** and **closure** that generalize what you have already seen in the real and complex analysis classes.

### Definition

A subset  $A$  of a topological space  $X$  is **closed** if set  $X \setminus A$  is **open**.

### Example

Both  $\emptyset$  and  $X$  are closed.

### Example

The subset  $[a, b]$  of  $\mathbb{R}$  under the standard topology is closed because its complement  $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, +\infty)$  is open. Also,  $[a, +\infty)$  and  $(-\infty, a]$  are closed, so that the terms “**closed interval**” and “**closed ray**” are justified.

## Example

Let  $Y = [0, 1] \cup (2, 3) \subset \mathbb{R}$  in the [subspace topology](#) (with  $\mathbb{R}$  under the [standard topology](#)). Then both  $[0, 1]$  and  $(2, 3)$  are open since  $[0, 1] = (-1/2, 3/2) \cap Y$  and  $(2, 3) = (2, 3) \cap Y$ .

Since in the topological space  $Y$ ,  $[0, 1]$  has  $(2, 3)$  as its complement and  $(2, 3)$  has  $[0, 1]$  as its complement, then  $[0, 1]$  and  $(2, 3)$  are also both closed. We shall encounter this type of example again when we address [connectedness](#).

## Theorem

*Let  $X$  be a topological space. Then the following conditions hold:*

- (1)**  $\emptyset$  and  $X$  are closed.
- (2)** Arbitrary intersections of closed sets are closed.
- (3)** Finite unions of closed sets are closed.

Instead of using open sets, one could just as well specify a topology on a space by giving a **collection of sets** (to be called “**closed sets**”) satisfying the three properties of this theorem. One could then **define open sets as the complements of closed sets** and proceed just as before.

## Proof

Since the complements of  $\emptyset$  and  $X$  are  $X$  and  $\emptyset$ , respectively, then by definition of closed, both  $\emptyset$  and  $X$  are closed (since  $X$  and  $\emptyset$  are open) and (1) follows.

Given a collection of closed sets  $\{A_\alpha\}_{\alpha \in J}$ , we have by DeMorgan's law (see Munkres' page 11),  $X \setminus \bigcap_{\alpha \in J} A_\alpha = \bigcup_{\alpha \in J} (X \setminus A_\alpha)$ . Since the sets  $X \setminus A_\alpha$  are open by definition, the right side of this equation is a union of open sets and so is open. Therefore the left hand side is open and so, by definition, its complement  $\bigcap_{\alpha \in J} A_\alpha$  is closed, as claimed in (2).

If  $A_i$  is closed for  $i = 1, 2, \dots, n$ , then again by DeMorgan's Law,  $X \setminus \bigcup_{i=1}^n A_i = \bigcap_{i=1}^n (X \setminus A_i)$ . The set on the right side is a finite intersection of open sets and is therefore open. So the left hand side is open and its complement,  $\bigcup_{i=1}^n A_i$ , is closed, as claimed in (3). ■

## Definition

If  $Y$  is a subspace of  $X$ , we say that a set  $A \subset Y$  is **closed in  $Y$**  (or **closed relative to  $Y$** ) if  $A \subset Y$  is **closed in the subspace topology of  $Y$**  (that is, if  $Y \setminus A$  is open in the subspace topology of  $Y$ ).

## Theorem

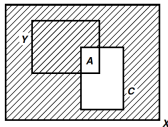
*Let  $Y$  be a subspace of  $X$ . Then a set  $A$  is closed in  $Y$  if and only if it equals the **intersection of a closed set of  $X$  with  $Y$** .*

## Proof

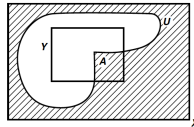
Suppose  $A = C \cap Y$  where  $C$  is closed in  $X$ . Then  $X \setminus C$  is open in  $X$  and so  $(X \setminus C) \cap Y$  is open in  $Y$  (by the definition of the subspace topology). But  $(X \setminus C) \cap Y = Y \setminus A$  (the complement of  $A$  is  $Y$ ), so  $Y \setminus A$  is open in  $Y$  and hence  $A$  is closed in  $Y$ .

Conversely, suppose that  $A$  is closed in  $Y$ . Then  $Y \setminus A$  is open in  $Y$  (by definition of “ $A$  is closed in  $Y$ ”). So, by definition of “open in  $Y \subset X$ ,” there is open  $U$  in  $X$  such that  $Y \setminus A = Y \cap U$ . Note that we have  $A = Y \cap (X \setminus U)$ . Now  $X \setminus U$  is closed in  $X$  since  $U$  is open in  $X$ . So  $A$  is the intersection of  $Y$  and a closed set  $X \setminus U$  of  $X$ . ■

Taken from [Munkres]



*Figure 17.1*



*Figure 17.2*

The proof of the following result is straightforward.

### Theorem

*Let  $Y$  be a subspace of  $X$ . If  $A$  is closed in  $Y$  and  $Y$  is closed in  $X$ , then  $A$  is closed in  $X$ .*

# Closure and interior of a set

## Definition

Given a subset  $A$  of a topological space  $X$ , the **interior** of  $A$ , denoted  $\text{Int}(A)$ , is the union of all open subsets contained in  $A$ . The **closure** of  $A$ , denoted  $\bar{A}$  (or sometimes  $\text{Cl}(A)$ ) is the intersection of all closed sets containing  $A$ .

The interior of  $A$  is open by part (2) of the definition of topology. The closure of  $A$  is closed by part (2) of Theorem on slide 47. Of course,

$$\text{Int}(A) \subset A \subset \bar{A}.$$

You may have been exposed the concept of an [interior point to a set of real numbers](#) in your analysis classes where such points are defined using an  $\varepsilon$  definition. However, we do not (necessarily) have a way to measure distance in a topological space, so we cannot take this approach. Things are about to get much stranger when we define a “limit point” of a set using only the topology and not using a distance-based concept of “closeness.” Once we consider metric spaces later on we will be back to a setting similar to your analysis experience.

## Lemma

*Let  $A$  be a subset of topological space  $X$ . Then  $A$  is open if and only if  $A = \text{Int}(A)$ .  $A$  is closed if and only if  $A = \bar{A}$ .*

## Proof

If  $A = \text{Int}(A)$  then, since  $\text{Int}(A)$  is open,  $A$  is open.

If  $A$  is open then, by the definition of  $\text{Int}(A)$  as the union of all open subsets contained in  $A$ , we have  $A \subset \text{Int}(A)$ . As commented above,  $\text{Int}(A) \subset A$  so if  $A$  is open then  $A = \text{Int}(A)$ .

If  $A = \bar{A}$  then, since  $\bar{A}$  is closed, then  $A$  is closed.

If  $A$  is closed then, by the definition of  $\bar{A}$  as the intersection of all closed sets containing  $A$ , we have  $\bar{A} \subset A$ . As commented above,  $A \subset \bar{A}$  so if  $A$  is closed then  $A = \bar{A}$ . ■

If  $Y$  is a subspace of  $X$  and  $A \subset Y$ , then  $A$  may have different closures in  $X$  and  $Y$ . So the symbol  $\bar{A}$  is ambiguous in this setting. Munkres reserves the symbol for the closure of  $A$  in  $X$ . The following result relates the closure of  $A$  in  $Y$  to the closure of  $A$  in  $X$ .

### Theorem

*Let  $Y$  be a subspace of  $X$ . Let  $A \subset Y$  and denote the closure of  $A$  in  $X$  as  $\bar{A}$ . Then the closure of  $A$  in  $Y$  equals  $\bar{A} \cap Y$ .*

### Proof

Let  $B$  denote the closure of  $A$  in  $Y$ . Since  $\bar{A}$  is closed in  $X$ , then  $\bar{A} \cap Y$  is closed in  $Y$  by a preceding Theorem (slide 49). Since  $\bar{A} \cap Y$  contains  $A$  (we are given  $A \subset Y$ ) and since, by definition,  $B$  equals the intersection of all closed subsets of  $Y$  containing  $A$ , so we must have  $B \subset \bar{A} \cap Y$ .

On the other hand,  $B$  is closed in  $Y$ . Hence by Theorem (slide 49),  $B = C \cap Y$  for some closed  $C$  in  $X$ . Then  $C$  is a closed set of  $X$  containing  $A$  (because  $A \subset B \subset C$ ). Now  $\bar{A}$  is the intersection of all closed sets in  $X$  containing  $A$ , so  $\bar{A} \subset C$ . Then  $\bar{A} \cap Y \subset C \cap Y = B$ . Therefore,  $\bar{A} \cap Y = B$ , as claimed. ■

## Definition

An open set  $U \subset X$  is a **neighborhood** of  $x \in X$  if  $x \in U$ .

## Theorem

Let  $A$  be a subset of the topological space  $X$ .

- (a) Then  $x \in \bar{A}$  if and only if **every neighborhood of  $x$  intersects  $A$**  (i.e. for each  $U$  open with  $x \in U$ , we have  $U \cap A \neq \emptyset$ ).
- (b) Supposing the topology of  $X$  is given a basis, then  $x \in \bar{A}$  if and only if every basis element  $B$  containing  $x$  intersects  $A$ .

## Example

Let  $X = \mathbb{R}$  under the standard topology. If  $A = (0, 1]$  then  $\bar{A} = [0, 1]$ . If  $B = \{1/n \mid n \in \mathbb{N}\}$  then  $\bar{B} = \{0\} \cup B$ . If  $C = \{0\} \cup (1, 2)$  then  $\bar{C} = \{0\} \cup [1, 2]$ .  $\overline{\mathbb{Q}} = \mathbb{R}$ ,  $\overline{\mathbb{Z}} = \mathbb{Z}$ ,  $\overline{\mathbb{R}_+} = \mathbb{R}_+ \cup \{0\}$  where  $\mathbb{R}_+ = \{x \in \mathbb{R} \mid x > 0\}$ .

# Limit points

Now we define a “limit point” of a set without an appeal to closeness, but only by using properties of the topology (which means we only use open sets).

## Definition

If  $A$  is a subset of topological space  $X$  and if  $x \in X$  then  $x$  is a **limit point** (or **cluster point** or **point of accumulation**) of  $A$  if every neighborhood of  $x$  intersects  $A$  in some point other than  $x$  itself. (i.e. for each  $U$  open with  $x \in U$ , we have  $(U \cap A) \setminus \{x\} \neq \emptyset$ ).

## Example

Consider the set  $A = \{0\} \cup (1, 2]$  in  $\mathbb{R}$  under the standard topology. Then  $\bar{A} = \{0\} \cup [1, 2]$ ,  $\text{int}(A) = (1, 2)$ , and the limit points of  $A$  are the points in  $[1, 2]$ . So  $0 \in A$  is a point of closure but not a limit point of  $A$ , and the points in  $[1, 2]$  are points of closure and limit points.

The following result gives a relationship between the closure of a set and its limit points.

### Theorem

Let  $A$  be a subset of the topological space  $X$ . Let  $A'$  be the *set of all limit points of  $A$* . Then

$$\bar{A} = A \cup A'.$$

### Proof

If  $x \in A'$  then every neighborhood of  $x$  intersects  $A$  in a point different from  $x$ . Therefore, by Theorem on slide 56,  $x$  belongs to  $\bar{A}$ . Hence  $A' \subset \bar{A}$ . Since  $A \subset \bar{A}$ , we have  $A \cup A' \subset \bar{A}$ .

Let  $x \in \bar{A}$ . If  $x \in A$ , then  $x \in A \cup A'$ . If  $x \notin A$  then, since  $x \in \bar{A}$ , every neighborhood  $U$  of  $x$  intersects  $A$ . Because  $x \notin A$  then  $U$  must intersect  $A$  in a point different from  $x$ . Then  $x \in A'$  so that  $x \in A \cup A'$ . Therefore,  $\bar{A} \subset A \cup A'$  and hence  $\bar{A} = A \cup A'$ , as claimed. ■

## Corollary

A subset of a topological space is *closed if and only if it contains all its limit points*.

## Proof

The set  $A$  is closed if and only if  $A = \bar{A}$  by Lemma on slide 54. By the preceding Theorem  $\bar{A} = A \cup A'$ , so  $A = \bar{A}$  if and only if  $A' \subset A$ . ■

$$A \text{ closed} \quad \Leftrightarrow \quad A = \bar{A} \quad \Leftrightarrow \quad A' \subset A$$

## Definition

Let  $x_1, x_2, \dots$  be a sequence of points in topological space  $X$ . The sequence **converges to  $x \in X$**  if **for every neighborhood  $U$  of  $x$  there exists  $N \in \mathbb{N}$  such that for all  $n \geq N$  we have  $x_n \in U$** . In this case,  $x$  is called a **limit** of the sequence.

The above definition is similar to the definition from calculus, but the “for all  $\varepsilon > 0$ ” part of is replaced with “**for every neighborhood**” here.

# Limit points and sequence limits

The terms **limit (of a sequence)** and **limit point** are often a source of confusion. On the real line, for instance, the constant sequence  $\{1, 1, \dots, 1, \dots\}$  is convergent with **limit** 1. But the set of points of this sequence is the set consisting of the single element 1 and is not a limit point of this set.

A sequence **may have a limit but without having a limit point** and **the set of points of a sequence may have a limit point without having a limit**.

We will see in the next chapter that for **metric spaces** **if a convergent sequence has infinitely many distinct points**, then **its limit is a limit point of the set of points of the sequence**.

## Example

Consider the topological space  $X = \{a, b, c\}$  with open set in  $\mathcal{T} = \{\emptyset, \{a, b\}, \{b\}, \{b, c\}, X\}$ . In this topology the sequence  $b, b, b, \dots$  **converges to  $b$**  (because every neighborhood of  $b$  contains all elements of the sequence), **converges to  $a$**  (because every neighborhood of  $a$ , the smallest such neighborhood being  $\{a, b\}$ , contains all elements of the sequence), and **converges to  $c$**  (because every neighborhood of  $c$ , the smallest such neighborhood being  $\{b, c\}$ , contains all elements of the sequence). **Weird!**

What about the sequences  $a, a, a, \dots$ ? the sequence  $c, c, c, \dots$ ?

This non-uniqueness of limits of sequences and the fact that singletons may not be closed sets, as motivation for the following definition.

## Definition

A topological space  $X$  is a **Hausdorff space** (or that satisfies the  **$T_2$  Axiom**) if for each pair of distinct points  $x_1, x_2 \in X$ , there exist neighborhoods  $U_1$  of  $x_1$  and  $U_2$  of  $x_2$  such that  $U_1 \cap U_2 = \emptyset$ .

## Theorem

*Every finite point set in a Hausdorff space  $X$  is closed. In particular, singletons form closed sets in a Hausdorff space.*

## Proof

Consider the set  $\{x_0\}$ . If  $x \in X$  where  $x \neq x_0$  then, since  $X$  is a Hausdorff space, there are disjoint neighborhoods  $U$  of  $x$  and  $V$  of  $x_0$ . Since  $U$  does not intersect  $\{x_0\}$ , by Theorem on slide 56, part (a),  $x$  is not in the closure of set  $\{x_0\}$ . Since  $x \neq x_0$  is an arbitrary element of  $X$ , the only points of closure of  $\{x_0\}$  is  $x_0$  itself and so by Lemma on slide 54,  $\{x_0\}$  is a closed set.

Now if we consider a finite point set, say  $\{x_0, x_1, \dots, x_n\}$ , then we simply write the set as  $\{x_0\} \cup \{x_1\} \cup \dots \cup \{x_n\}$ , observe that each  $\{x_i\}$  is closed, and apply part (3) of the Theorem on slide 47. ■

## Theorem

Let  $X$  be a space satisfying the  $T_1$  Axiom (namely, that *all finite point sets are closed*). Let  $A$  be a subset of  $X$ . Then  $x$  is a limit point of  $A$  if and only if *every neighborhood of  $x$  contains infinitely many points of  $A$* .

Recall:  $x$  is a limit point of  $A$  if every neighborhood of  $x$  intersects  $A$  in some point other than  $x$  itself.

## Proof

Suppose every neighborhood of  $x$  intersects  $A$  in infinitely many points. Then every neighborhood of  $x$  intersects set  $A$  at a point other than  $x$  and so (by definition)  $x$  is a limit point of  $A$ .

Conversely, suppose that  $x$  is a limit point of  $A$ . ASSUME some neighborhood  $U$  of  $x$  intersects  $A$  in only finitely many points. Then  $U$  also intersects  $A \setminus \{x\}$  in finitely many points, say  $\{x_1, x_2, \dots, x_m\} = U \cap (A \setminus \{x\})$ . The set  $X \setminus \{x_1, x_2, \dots, x_m\}$  is open in  $X$  since by  $T_1$  Axiom  $\{x_1, x_2, \dots, x_m\}$  is closed. Then  $U \cap (X \setminus \{x_1, x_2, \dots, x_m\})$  is a neighborhood of  $x$  that does not intersect the set  $A \setminus \{x\}$ . But this CONTRADICTS the hypothesis that  $x$  is a limit point of  $x$ . So the assumption that  $U$  intersects  $A$  in finitely many points is false. That is, any neighborhood of  $x$  must intersect  $A$  in infinitely many points. ■

## Theorem

If  $X$  is a Hausdorff space, then a sequence of points of  $X$  *converges to at most one point of  $X$* .

## Proof

Let  $x_n$  be a sequence of points of  $X$  that converges to  $x$  and let  $y \neq x$ . We shall show that the sequence does not converge to  $y$ . Let  $U$  and  $V$  be disjoint neighborhoods of  $x$  and  $y$ , respectively. Since  $U$  is a neighborhood of  $x$ , then there is  $N_1 \in \mathbb{N}$  such that  $x_n \in U$  for all  $n \geq N_1$ . So there is no  $N_2 \in \mathbb{N}$  such that for  $n \geq N_2$  we have  $x_n \in V$  (since for  $n \geq N_1$ ,  $x_n \in U$  and  $U \cap V = \emptyset$ ). That is,  $x_n$  does not converge to  $y \neq x$  and  $x_n$  converges to at most one point in  $X$ . ■

# Outline

- 1 Topological spaces
- 2 Basis of a topology
- 3 The order topology
- 4 The product topology
- 5 The subspace topology
- 6 Closed sets and limit points
- 7 References**

# References I



James Munkres. Topology. Pearson, second edition, 2014.